# Project 5 — Looking Ahead

**Prediction with a Linear Model**

# Project 5: Stone prediction challenge

**Total points:** 40

Your goal is to build, evaluate, and use a simple linear regression model to predict the number of stones in Rosanna's cup given it's total weight.

This project is also practice for doing statistics on your own after this course. As a result, there is no sample notebook for this project: you will show that you can open your own notebook, write the code from scratch, and produce a short report. Use the demos, the R reference, the internet, and AI tools as needed to work out the code, just as you will in real work after the course is over.

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## What you will do

1. Collect data with the Rosanna's Stone Jar sampler. Choose a number of stones, weigh that random sample, and record your data. Repeat as often as you like.
2. Organize your data in a spreadsheet (Google Sheets or Excel). A tidy format puts variables in columns and observations per unit in rows.
3. Open Google Colab, switch the runtime to R, and save a copy to your own Google Drive. Then write code from scratch to:
   - load the libraries you need,
   - load your data (read directly from your Google Sheet, or download it as a `.csv` and upload it with the Colab file manager as shown in a demo),
   - check that your data loaded correctly and that the variable types are correct,
   - explore the data with summaries and a scatterplot,
   - fit a simple linear regression model,
   - assess the model,
   - make a prediction with a 95% prediction interval for the number of stones in Rosanna's cup given that it's total weight is 209.27 grams.

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## Collecting your data

- Use the Rosanna's Stone Jar sampler to weigh random samples of stones.
- Record each pair (number of stones, total weight in grams). The total weight already includes the cup and lid.
- Keep your spreadsheet tidy: one row per sample and informative column names.

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## Your notebook

Write your notebook so it runs top to bottom, in this order:

1. **Load and check the data.** Load your spreadsheet, then show the first rows, the variable names, the number of rows, and the type of each variable.
2. **Explore.** Report numerical summaries of the response and predictor (center and spread, plus the minimum and maximum of the predictor, since these set the valid range for prediction), report the Pearson correlation, and make

a labeled scatterplot. Comment on form, direction, and strength.

3. **Fit.** Fit a simple linear regression model thinking carefully about which variable should be predicted ($y$) and which variable should be used to predict from ($x$). Report the prediction equation and interpret the slope and intercept in context, with units.
4. **Assess.** Evaluate the model (for example, whether a straight line is appropriate, the residuals, and $R^2$ or MSE).
5. **Predict.** Predict the number of stones in Rosanna's cup (it's total weight is 209.27 g) and report a 95% prediction interval for this single number (`predict` with `interval = "prediction"`).

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# What to turn in

1. **Analysis notebook** (`.ipynb`) that runs top to bottom (5 points).
2. **Data spreadsheet** (a `.csv`, or a link to your Google Sheet, 5 points).
3. **One-page report** (PDF, 8.5 by 11 in.) with the sections below (20 points).

## One-page report sections

Use short, clear headings:

1. **Introduction.** The problem context and a clear problem statement: what you are predicting and what you are predicting it from.
2. **Methods: Data Collection.** How you collected the data (for example, "I used the stone jar sampler to weigh random samples of stones, recording the number of stones and the total weight for each sample"). Include the number of times you weighted samples.
3. **Methods: Data Analysis.** The steps you took to build, assessed and use a prediction model. (for example, explored the relationship with a scatterplot and correlation, fit a simple linear regression of number of stones on total weight, and used `predict` for a 95% prediction interval). How you assessed model performance (e.g. scatterplot looked linear, MSE or $R^2$ value).
4. **Results.** The labeled scatterplot, the Pearson correlation, the prediction equation with interpreted slope and intercept, and the range of valid inputs with units. Give MSE or $R^2$. (note: the methods section is where you describe what you did, the results section is what you got when you followed the methods section).
5. **Conclusion.** Your best prediction for the number of stones in Rosanna's cup, with its 95% prediction interval, stated in context. State how you could get a more precise (narrower) prediction interval in the future and whether or not you think your prediction might be biased. (Hint: what would happen if you collected data from more samples? Does random sampling influence bias?).

## Report rubric (20 points)

| Section | Points | Full credit when |
|---|---|---|
| Introduction | 3 | The problem context is given and the problem statement clearly says what you are predicting and what you are predicting it from. |
| Methods: Data Collection | 3 | The collection process is described clearly enough to replicate, including how the sampler was used and how many samples you weighed. |
| Methods: Data Analysis | 4 | Describes how you built, assessed, and used the model (scatterplot and correlation, `lm` of number on total weight, `predict` for a 95% prediction interval) and how you judged model performance (linear scatterplot, MSE or $R^2$). |
| Results | 6 | Includes a labeled scatterplot, the Pearson correlation, the prediction equation with slope and intercept interpreted in context with units, the valid range of inputs, and the MSE or $R^2$. |
| Conclusion | 4 | States the best prediction with its 95% prediction interval in context, explains how to obtain a narrower interval, and discusses whether the prediction may be biased. |